Foundation of Blockchain Technology (IS501)
Pre-Reading Material – Lecture 33

Verifiable Random Function (VRF)-I

Estimated Reading Time: 15–20 minutes

Introduction

A Verifiable Random Function (VRF) is a cryptographic function that produces a seemingly random output along with a proof that allows others to verify the correctness of that output.

In blockchain technology, randomness is useful for selecting validators, assigning tasks, conducting lotteries, and supporting consensus mechanisms. However, ordinary random-number generation may be difficult to trust in a decentralized environment because participants might manipulate or selectively reveal random values.

A VRF helps address this challenge by combining randomness with cryptographic verifiability.

What Is a Verifiable Random Function?

A VRF is a cryptographic mechanism that takes a private key and an input and produces:

- Output: A pseudorandom-looking value derived from the input and private key.
- Proof: A cryptographic value that allows others to verify that the output was generated correctly.

A verifier uses the corresponding public key, the input, the output, and the proof to check validity.

The same private key and input produce the same VRF output in a properly specified deterministic VRF, while different inputs generally produce different outputs.

Main Components of a VRF

The main components are:

Private key: A secret key used to generate the VRF output and proof.

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Public key: A key associated with the private key and used to verify the result.

Input: The data supplied to the VRF, such as a block identifier, round number, or protocol-specific value.

VRF output: The pseudorandom-looking result generated from the input and private key.

Proof: Cryptographic evidence that the output is valid for the specified input and public key.

How Does a VRF Work?

A VRF generally involves three operations:

A. Key Generation

A participant generates a private key and its corresponding public key. The private key must be protected, while the public key can be made available for verification.

B. Output and Proof Generation

The participant supplies an input to the VRF and uses the private key to generate an output and its associated proof.

C. Verification

Another participant uses the public key, input, output, and proof to verify that the output is valid.

If verification succeeds, the verifier gains confidence that the output was generated correctly by the VRF associated with that public key.

Why Is Verifiable Randomness Important?

In decentralized systems, random selection must be difficult to manipulate and easy to verify.

For example, imagine a blockchain network that needs to select a participant to propose the next block. If a participant can freely choose a favorable random number, the selection process may become unfair.

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A VRF can provide a verifiable output that a protocol uses to determine eligibility. Other participants can verify the result without learning the private key.

However, a VRF alone does not guarantee that an entire selection protocol is fair. Fairness also depends on input selection, eligibility rules, incentives, and protections against manipulation.

Properties of a VRF

- Verifiability: Others can check the validity of the output using the proof.
- Pseudorandomness: The output appears random to parties lacking the relevant secret information, subject to the VRF's security assumptions.
- Uniqueness: For a given public key and input, a secure VRF is designed to prevent an adversary from producing two different valid outputs.
- Deterministic evaluation: The same private key and input yield the same output in a deterministic VRF.
- Private-key protection: Verification does not require revealing the private key.

Applications of VRFs in Blockchain

VRFs can be used in:

- Validator or leader selection.
- Randomized committee selection.
- Lottery-based blockchain protocols.
- Random assignment of tasks.
- Certain consensus mechanisms that require verifiable eligibility.

For example, Algorand uses cryptographic sortition with VRF-based proofs to support the selection of participants for consensus activities.

Summary

A Verifiable Random Function combines pseudorandom output generation with cryptographic proof. It allows a participant to produce an output using a private key and enables other participants to verify that output using the corresponding public key. VRFs are useful in blockchain applications where randomness must be verifiable and resistant to manipulation.

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